BUSINESS PROPOSAL

UMHackathon 2026

umhackathon@um.edu.my

Email: umhackathon@um.edu.my

1. Executive Summary:

InvenIQ is an AI-powered conversational inventory optimization platform designed for small and medium-sized enterprises (SMEs) in the retail sector. Many SMEs rely on manual processes or intuition when managing inventory, resulting in overstocking, stock shortages, and reduced profitability.

InvenIQ addresses this issue by transforming fragmented business data into actionable insights through a simple chat-based interface. By leveraging Z AI (GLM), the system combines structured data such as sales records with unstructured inputs like supplier messages, weather conditions, and local events to generate context-aware inventory strategies.

Targeting SME owners, retail managers, and e-commerce sellers, InvenIQ provides an accessible and cost-effective alternative to complex enterprise systems. The platform aims to improve operational efficiency, reduce waste, and maximize revenue for businesses.

2. Product Overview / Solution

Key Features: Highlight key features of InvenIQ

- Conversational AI interface for simple, chat-based interaction
- Smart and automated inventory recommendations based on historical sales data
- Multi-format input (CSV, manual entry, unstructured text)
- Integration of real-time external signals (e.g. news and weather trends)
- Explainable AI Recommendations that accompanied by a plain-language rationale to support decision-making
- Strategy export feature for easy implementation (e.g., PDF reports)

Technology Used: Highlight the technology used in InvenIQ

- Z AI (GLM) as the core reasoning and natural language processing engine
- API integration for external data such as weather and local signals
- Cloud-based deployment (Render) for scalability and accessibility

3.1 Problem Landscape

SMEs in Malaysia often operate with limited access to advanced inventory management tools and rely heavily on manual decision-making. This leads to inefficient stock planning, including overstocking which ties up capital and understocking which results in lost sales.

Existing solutions are typically expensive and require technical expertise, making them unsuitable for small businesses. As a result, many SMEs are unable to fully utilize their data to improve operational efficiency and profitability.

3.2 Market Niche

InvenIQ targets SME business owners, retail managers, and e-commerce sellers who require simple and practical tools for inventory decision-making. These users often lack formal financial or technical training and prefer intuitive solutions.

The platform focuses on businesses that currently rely on manual tracking or basic tools such as spreadsheets, offering them a pathway to digital transformation without high costs.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	All SMEs and e-commerce sellers in Malaysia (~1.5 million users × RM588/year)	RM 882 million
SAM	Digitally active SMEs and sellers (~40% of TAM, ~600,000 users × RM588/year	RM 352.8 million
SOM	Realistic market capture within **18 months** (Targeting ~8,000–10,000 urban users × RM 588/year))	RM 4.7 – 5.8 million

*For SAM, we are excluding service-based SMEs (like hair salons, consulting firms, or laundromats) that do not require complex inventory forecasting. The 40% represents the segment of SMEs that are already "digitally active"

*By setting SOM at **8,000 to 10,000 users**, we assume that our 18-month goal is to reach the Year 2 milestone in our financial table.

3.4 Market Dynamics & Placement:

There were several key dynamic that shape InvenIQ's entry and growth to this market:

- **Smartphone Penetration:** Smartphone adoption in Malaysia exceeds 80%, enabling SMEs to easily access browser-based solutions like InvenIQ via mobile devices without requiring a dedicated app.
- **SME Digitalization Push:** The Malaysian government actively promotes SME digital transformation through initiatives such as grants, subsidies, and digital adoption programs. These programmes encourage SMEs to adopt digital tools, creating a supportive environment for inventory optimization solutions like InvenIQ.
- **Fragmented Competitor Landscape:** Existing solutions are either too complex and expensive or lack AI capabilities. InvenIQ fills this gap by offering an affordable, conversational, and locally relevant inventory intelligence tool for SMEs.
- **Digitalization of Owners:** SME owners are increasingly shifting toward digital platforms to improve efficiency and reduce reliance on traditional manual processes.

4. Value Proposition & Competitive Advantage

InvenIQ simplifies inventory decision-making by transforming complex sales and operational data into clear, actionable recommendations through a conversational interface. Unlike traditional systems that rely on dashboards and technical expertise, InvenIQ allows SME owners to make decisions using simple natural language.

The platform integrates structured data (e.g., sales records) with unstructured inputs (e.g., supplier messages, weather, and local trends), enabling more context-aware inventory insights. This helps SMEs respond faster to demand changes and reduce stock inefficiencies.

A key competitive advantage is explainable AI, where every recommendation is accompanied by a clear reasoning, increasing trust and usability among non-technical users.

Overall, InvenIQ delivers an affordable, accessible, and intelligent inventory system designed specifically for SMEs, bridging the gap between advanced AI capabilities and real-world business needs.

5. Competitor Analysis (Porter's Five Forces)

Force	***Intensity***	***Assessment***
Competitive Rivalry	High	The space is crowded with high-end AI solutions like Netstock and Blue Yonder. However, these are often "Dashboard-First," requiring significant setup. InvenIQ competes by being "Conversation-First," reducing the barrier to entry for non-technical SME owners.
Threat of New Entrants	High	With the democratisation of LLMs, new AI "wrappers" appear weekly. InvenIQ's defense is its Multi-Modal Local Signal Integration (Malaysian weather, local news via Z.AI GLM-5.1) which generic global tools often overlook.
Bargaining Power of Buyers	Medium	While Malaysian SMEs are price-sensitive, the high cost of enterprise tools (like IBM Watson or Kinaxis) makes InvenIQ an attractive, high-value alternative, increasing user stickiness.
Bargaining Power of Suppliers	High	InvenIQ is strategically tied to the Z.AI GLM-5.1 ecosystem. This is a deliberate "all-in" move to leverage the model's superior reasoning for

		economic empowerment, as required by the hackathon domain.
Threat of Substitutes	Low	Global platforms like Netstock require integration with expensive ERPs (Sage/NetSuite). For a Malaysian SME using only Excel or manual records, InvenIQ has no direct "affordable AI" substitute.

Competitor Comparison Table:

Competitor	***Strength***	***Weakness***	***InvenIQ's Advantage***
Netstock / Unframe	Excellent "Actionable Recommendations" and "Daily Brief" planners for established SMBs.	Requires existing high-end ERP data (Sage/NetSuite) to function; high monthly overhead.	Data Agnostic: InvenIQ can reason over a simple raw CSV or even a WhatsApp text message from a supplier using Z.AI GLM's unstructured data capability.
Blue Yonder / IBM Watson	"Gold Standard" for autonomous supply chains and predictive demand planning.	Designed for Global Enterprises; "Black Box" decisions that are hard for small business owners to understand.	Explainable AI: InvenIQ doesn't just give a number; it provides a natural language explanation (e.g., "I suggest stocking up because the heatwave next week will likely drive beverage demand by 20%").
Leafio / Nextail	Strong "What-if" analysis and alert-driven replenishment for retail.	Niche focus (e.g., Fashion); lacks integration with real-time local external signals (Malaysian news/events).	Hyper-Local Intelligence: Integrated with live Malaysian news and weather signals to provide context that global retail tools miss.

6.1. Customer Segmentation

1. **Micro-SMEs & Retailers:** Small shop owners (Grocers, Boutiques, Florists) in urban and sub-urban Malaysia who currently rely on manual "gut-feeling" for inventory but suffer from stockouts or wastage.
2. **F&B Operators:** Cafes and "Warungs" whose demand is highly sensitive to external factors like weather and local events—perfect for InvenIQ's context-aware reasoning.
3. **E-commerce Vendors:** Shopee/Lazada sellers managing high-frequency stock across multiple channels who need a simplified "Daily Brief" to avoid over-ordering.

6.2. Revenue Stream

Primary: SaaS Subscription (B2C / Micro-SMEs)

- **Starter Tier (RM 0 / Freemium):** Basic inventory tracking (up to 50 SKUs) and manual CSV uploads. Acts as our customer acquisition funnel.
- **Professional Tier (RM 49/month or RM 499/year):** Full **Z.AI GLM-powered reasoning**, real-time weather/news signal integration, and unlimited CSV uploads. *(Note: This aligns perfectly with our RM 600 ARPU calculation in the financials).*

Secondary: API Integration Licensing (B2B / POS Vendors)

- **Platform License (RM 999/month per POS Provider):** Charging a flat recurring fee to local Point of Sale software companies (e.g., Slurp!, StoreHub) to embed InvenIQ's "Decision Engine" directly into their cash registers.
- **Usage Fee:** RM 0.05 per API call once the base token limit is exceeded, creating a scalable revenue ceiling.

Tertiary: Data Intelligence Reports (Enterprise / FMCG)

- **Quarterly Trend Reports (RM 3,500 per report):** Custom-generated, anonymized market insights for specific sectors (e.g., "Q3 Beverage Demand Shifts in Klang Valley").
- **Enterprise Data Dashboard (RM 25,000/year):** Direct API access to macro-level trends for large distributors and supply chain giants.

6.3. Cost Structure

Cost Drivers	Cost Type
API Inference & Token Usage: Costs associated with Z.AI's GLM-5.1. (Includes logic for Token Efficiency to ensure manageable operational costs).	Variable Cost
Cloud Infrastructure & Maintenance: Render cloud hosting (Web Services for long-running AI processes), database maintenance, and frontend hosting.	Fixed Cost
R&D & Product Development: Continuous refinement of the "Contextual Signal" logic and prompt engineering for the Z.AI engine.	Fixed Cost
Marketing & SME Outreach: Costs for onboarding local vendors and partnering with SME associations (MDEC).	Overhead Cost

6.4. Key Partners

Internal (Ecosystem)	External
Z.AI: The core provider of the GLM reasoning engine and API infrastructure.	MDEC & SME Corp: For national-level digitalization grants and user outreach.
InvenIQ Tech Team: Managing the CI/CD pipeline and Z.AI model integration.	Weather & News APIs: Providers for wttr.in and search signal data to feed into the GLM.
Local Payment Gateways: (e.g., ToyyibPay/Billplz) for subscription handling.	Logistics/Courier Partners: Potential future integration for automated restocking.

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
API Rate Limits	High	Medium	Implement Token Efficiency strategies (caching) and graceful degradation to basic logic if the Z.AI API is under heavy load.
Data Privacy	High	Low	End-to-end encryption of user CSVs and ensuring no sensitive PII is sent

			to the Z.AI model during reasoning.
Forecast Inaccuracy	High	Medium	Ground the AI in RAG (Retrieval-Augmented Generation) using the user's actual historical sales data to prevent "hallucinations."
Low SME Adoption	Medium	High	Focus on a Conversational "Chat" UI that feels like WhatsApp, leveraging Z.AI’s natural language capabilities to remove the "fear of technology."

7. Implementation & Financials

7.1 Go-to-Market Strategy

InvenIQ will adopt a **"Cluster-Based Digitalization"** strategy. We will focus on high-density SME hubs starting with the Klang Valley (Kuala Lumpur/Selangor), moving into Penang and Johor Bahru.

- **Strategic Partnerships**: Collaborate with **MDEC** and **SME Corp Malaysia** to list InvenIQ as a recommended solution under the *SME Digitalisation Grant*.
- **Product-Led Growth**: Offer a **"30-Day Decision Intelligence Challenge"** where SMEs use InvenIQ for free. We will quantify their "Reduction in Stockouts" or "Wastage Savings" during the trial to convert them into paid subscribers.
- **Community Outreach**: Partner with local retail associations (e.g., Malaysia Retailers Association) for physical workshops demonstrating GLM-powered forecasting.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Soft Launch (Beta)	Pilot with 100 select retailers in Klang Valley; ground-truth GLM accuracy.	Month 1-4	1,000 Recommendations/day
Expansion Phase	Regional rollout to Penang and Johor; launch "WhatsApp Signal" integration.	Month 4-12	5,000 Active Subscriptions
AI Optimization	Refine GLM token efficiency; introduce Multi-Echelon supply chain reasoning.	Month 12-18	Reach Net Breakeven
TOTAL	RM 2.5M Seed Investment	18 Months	15% Market Share in F&B/Retail

7.3 Revenue Projections

Metric	*Year 1*	*Year 2*	*Year 3*	*Year 4*	*Year 5*
Active SME Subscribers	1,200	8,500	25,000	55,000	110,000
Avg. Revenue Per User (RM/yr)	600	620	650	700	800
SaaS Subscription Revenue	RM 0.72M	RM 5.27M	RM 16.25M	RM 38.5M	RM 88.0M

Enterprise API Licensing	RM 0.10M	RM 1.03M	RM 4.75M	RM 10.7M	RM 15.0M
Total Revenue	RM 0.82M	RM 6.30M	RM 21.0M	RM 49.2M	RM 103M
Net Position (EBITDA)	RM (1.2M)	RM 0.5M	RM 8.4M	RM 22.0M	RM 52.0M
Active SME Subscribers	1,200	8,500	25,000	55,000	110,000

7.4 Key Financial Milestones

- **Month 6**: Secure MDEC Digitalization Grant partnership.
- **Month 14**: Reach **Operationally Cashflow Positive** (Revenue exceeds API costs + overhead).
- **Year 3**: Achieve **RM 20M+ Revenue**; initiate feasibility for Southeast Asia (SEA) expansion (Indonesia/Vietnam).
- **Year 5**: InvenIQ becomes the primary Decision Intelligence layer for **10% of all registered Malaysian SMEs**.

7.5 Cost Breakdown & Funding Strategy

To execute an agile yet sustainable 18-month Go-To-Market strategy, InvenIQ is raising a seed round of **RM 1.5 Million**. We have optimized our operational structure by combining key departments, allowing us to maintain a low burn rate while scaling the **Z.AI Decision Engine** ahead of the 2026 E-Invoicing mandate.

7.5.1 Capital Allocation (18-Month Runway)

By adopting a "Founder-Led Sales" model and automating our DevOps, we ensure maximum capital efficiency over an extended 1.5-year runway.

Department / Expenditure Category	Allocation (%)	Amount (RM)	Lean Operational Strategy & Manpower
Product & AI Engineering	50%	750,000	2x Core Developers (1 AI/Backend, 1 Full-Stack). Covers Vercel cloud infrastructure and strictly capped Z.AI API Token Inference Costs.
Growth, Marketing &	50%	750,000	Led by the CEO (Acting

Customer Success			Growth Lead) alongside a 2-person Marketing team. To maximize efficiency, the Marketing team directly handles Customer Success, combining user acquisition with zero-touch onboarding and prompt-support.

7.5.2 Unit Economics & The Break-Even Target

With our consolidated team structure, our annualized fixed burn rate is kept strictly at approximately **RM 1.0 Million** (RM 1.5M spread over 18 months).

- **Average Revenue Per User (ARPU):** RM 600 / year
- **Variable Cost Per User:** RM 150 / year *(Strictly enforced via **Z.AI Token Efficiency** limits, caching localized queries, and prompt-size restrictions).*
- **Contribution Margin:** RM 450 per user (75% Gross Profit).

The Break-Even Calculation:

Break-Even Volume = Annualized Fixed Costs $\div$ Contribution Margin

Break-Even Volume = RM 1,000,000 $\div$ RM 450 = **2,222 Active Subscribers**

7.5.3 The Path to Profitability

Operating with an 18-month runway significantly reduces the pressure to "Blitzscale," allowing for high-quality, sustainable growth.

Based on our Revenue Projections (Section 7.3), InvenIQ will secure **1,200 users by the end of Year 1**. With the CEO directly leading growth and the marketing team managing retention, we project a very low churn rate.

We will cross the 2,222 user break-even threshold by **Month 15**. Because our seed capital covers 18 months, we achieve full operational break-even *before* the funding runs out. This means InvenIQ becomes a **cashflow-positive, self-sustaining AI enterprise** without the immediate desperation of raising a Series A round.